INCH-POUND
MIL-PRF-39002/1G
w/Amendment 2
12 October 2018
SUPERSEDING
MIL-PRF-39002/1G
w/Amendment 1
27 October 2015

PERFORMANCE SPECIFICATION

RESISTOR, VARIABLE, WIREWOUND, SEMI-PRECISION, STYLE RK09

This specification is approved for use by all Departments
And Agencies of the Departments of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39002.

Inactive for New Design,
12 October 2018
Use DLA Land and Maritime
Drawing 18001

1. SCOPE

1.1 Scope. This specification covers the associated requirements for a variable, wirewound, semi-precision resistors. These resistors are capable of full load operation at a maximum ambient temperature of 85°C, and suitable for continuous operation. These resistors have a tolerance of ±5 percent.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number taken from table I. The PIN is in the following form:

M39002/01	-	0001
Performance specification number		Coded dash number

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information are as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A
FSC 5905

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-39002 - Resistor, Variable, Wirewound, Semi-Precision, General Specification for.

(Copies of these documents are available online at http://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-39002.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3 Power rating. The power rating shall be 1.5 watts based on full load operation at an ambient temperature of 85°C.

3.4 Resistance value. Resistance value shall be in accordance with table I.

3.5 Shaft length. Shaft length shall be in accordance with table I.

3.6 Shaft style. Shaft style shall be in accordance with table I.

3.7 Bushing. The bushing shall be in accordance with table I.

3.8 Nominal resolution. Nominal resolution shall be in accordance with table I.

3.9 Rated working voltage. Rated working voltage shall be in accordance with table I.

3.10 Mechanical travel. The mechanical travel shall be $300^{O} \pm 5^{O}$.

3.11 Actual effective electrical travel. The actual effective electrical travel shall be $300^{O} \pm 5^{O}$.

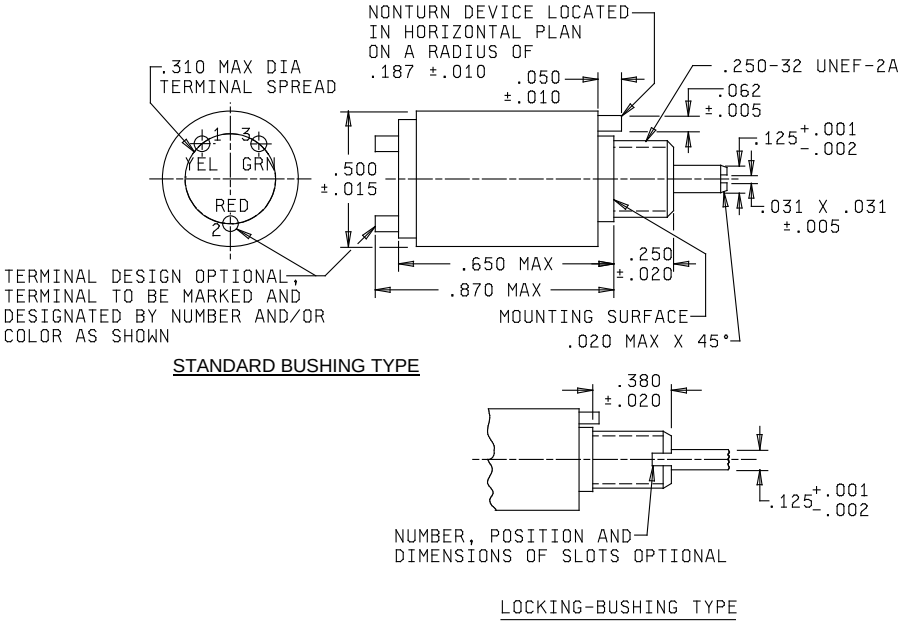
3.12 End resistance. The end resistance shall not exceed 3 percent of the nominal total resistance value or 1 ohm, whichever is greater.

3.13 Electrical output. The electrical output (in terms of percent of applied voltage) is linear with respect to the angular position of the contact arm.

3.14 Hardware. The mounting nut supplied shall be 0.0625 inch (1.588 mm) thick and shall measure 0.3125 inch (7.938 mm) across the hexagonal flats. For locking bushings, the locking nut shall be 0.125 inch (3.18 mm) thick and shall measure 0.3125 inch (7.938 mm) across the hexagonal flats. Thread size shall be 1/4-32NEF-2B. An internal-tooth lock washer shall be supplied and when mounted, shall have a maximum thickness of approximately 0.045 inch (1.14 mm). If retainer rings are used, they shall be not thicker than 0.032 inch (0.81 mm).

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3.15 Pure tin. The use of pure tin, as an underplate or final finish is prohibited both internally and externally. Tin content of resistor components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.4).



Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.010	0.25	.031	0.79	.125	3.18	.310	7.87	.650	16.51
.002	0.05	.015	0.38	.050	1.27	.187	4.75	.380	9.65	.870	22.10
.005	0.13	.020	0.51	.062	1.57	.250	6.35	.500	12.70		

NOTE:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style RK09 resistor.

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TABLE I. Style RK09 resistor.

Locking Bushing						
Dash number	Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal
0001	RK09UACS100		10	.66	3.8	
0002	RK09UACS150		15	.88	4.9	
0003	RK09UACS200		20	.89	5.7	
0004	RK09UACS250		25	.80	6.2	
0005	RK09UACS350		35	.72	7.3	
0006	RK09UACS500		50	.65	8.6	
0007	RK09UACS750		75	.55	10	
0008	RK09UACS101		100	.68	12	
0009	RK09UACS151		150	.41	14	
0010	RK09UACS201		200	.55	17.4	
0011	RK09UACS251		250	.39	19	
0012	RK09UACS351		350	.36	23	
0013	RK09UACS501		500	.42	27	
0014	RK09UACS751		750	.35	32	
0015	RK09UACS102	.500	1,000	.34	39	S
0016	RK09UACS152		1,500	.31	46	
0017	RK09UACS202		2,000	.23	55	
0018	RK09UACS252		2,500	.21	61	
0019	RK09UACS352		3,500	.21	73	
0020	RK09UACS502		5,000	.17	86	
0021	RK09UACS752		7,500	.21	104	
0022	RK09UACS103		10,000	.15	122	
0023	RK09UACS153		15,000	.13	150	
0024	RK09UACS203		20,000 3/	.12	173	
0025	RK09UACS253		25,000	.114	194	
0026	RK09UACS353		35,000	.102	229	
0027	RK09UACS503		50,000	.080	272	
0190	RK09UACS753		75,000 4/	.060	290	
0191	RK09UACS104		100,000 4/	.045	300	

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Standard bushing						
Dash number	Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal
0028	RK09SACS100		10	.66	3.8	
0029	RK09SACS150		15	.88	4.9	
0030	RK09SACS200		20	.89	5.7	
0031	RK09SACS250		25	.80	6.2	
0032	RK09SACS350		35	.72	7.3	
0033	RK09SACS500		50	.65	8.6	
0034	RK09SACS750		75	.55	10	
0035	RK09SACS101		100	.68	12	
0036	RK09SACS151		150	.41	14	
0037	RK09SACS201		200	.55	17.4	
0038	RK09SACS251		250	.39	19	
0039	RK09SACS351		350	.36	23	
0040	RK09SACS501		500	.42	27	
0041	RK09SACS751		750	.35	32	
0042	RK09SACS102	.500	1,000	.34	39	S
0043	RK09SACS152		1,500	.31	46	
0044	RK09SACS202		2,000 3/	.23	55	
0045	RK09SACS252		2,500	.21	61	
0046	RK09SACS352		3,500	.21	73	
0047	RK09SACS502		5,000	.17	86	
0048	RK09SACS752		7,500	.21	104	
0049	RK09SACS103		10,000	.15	122	
0050	RK09SACS153		15,000	.13	150	
0051	RK09SACS203		20,000	.12	173	
0052	RK09SACS253		25,000	.114	194	
0053	RK09SACS353		35,000	.102	229	
0054	RK09SACS503		50,000	.080	272	
0192	RK09SACS753		75,000 4/	.060	290	
0193	RK09SACS104		100,000 4/	.045	300	

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Standard bushing						
Dash number	Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal
0055	RK09SBCS100		10	.66	3.8	
0056	RK09SBCS150		15	.88	4.9	
0057	RK09SBCS200		20	.89	5.7	
0058	RK09SBCS250		25	.80	6.2	
0059	RK09SBCS350		35	.72	7.3	
0060	RK09SBCS500		50	.65	8.6	
0061	RK09SBCS750		75	.55	10	
0062	RK09SBCS101		100	.68	12	
0063	RK09SBCS151		150	.41	14	
0064	RK09SBCS201		200	.55	17.4	
0065	RK09SBCS251		250	.39	19	
0066	RK09SBCS351		350	.36	23	
0067	RK09SBCS501		500	.42	27	
0068	RK09SBCS751		750	.35	32	
0069	RK09SBCS102	.625	1,000	.34	39	S
0070	RK09SBCS152		1,500	.31	46	
0071	RK09SBCS202		2,000	.23	55	
0072	RK09SBCS252		2,500	.21	61	
0073	RK09SBCS352		3,500	.21	73	
0074	RK09SBCS502		5,000	.17	86	
0075	RK09SBCS752		7,500	.21	104	
0076	RK09SBCS103		10,000	.15	122	
0077	RK09SBCS153		15,000	.13	150	
0078	RK09SBCS203		20,000 3/	.12	173	
0079	RK09SBCS253		25,000	.114	194	
0080	RK09SBCS353		35,000	.102	229	
0081	RK09SBCS503		50,000	.080	272	
0194	RK09SBCS753		75,000 4/	.060	290	
0195	RK09SBCS104		100,000 4/	.045	300	

See footnotes at end of table.

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Locking bushing						
Dash number	Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal
0082	RK09UDCS100		10	.66	3.8	
0083	RK09UDCS150		15	.88	4.9	
0084	RK09UDCS200		20	.89	5.7	
0085	RK09UDCS250		25	.80	6.2	
0086	RK09UDCS350		35	.72	7.3	
0087	RK09UDCS500		50	.65	8.6	
0088	RK09UDCS750		75	.55	10	
0089	RK09UDCS101		100	.68	12	
0090	RK09UDCS151		150	.41	14	
0091	RK09UDCS201		200	.55	17.4	
0092	RK09UDCS251		250	.39	19	
0093	RK09UDCS351		350	.36	23	
0094	RK09UDCS501		500	.42	27	
0095	RK09UDCS751		750	.35	32	
0096	RK09UDCS102	0.875	1,000	.34	39	S
0097	RK09UDCS152		1,500	.31	46	
0098	RK09UDCS202		2,000	.23	55	
0099	RK09UDCS252		2,500	.21	61	
0100	RK09UDCS352		3,500	.21	73	
0101	RK09UDCS502		5,000	.17	86	
0102	RK09UDCS752		7,500	.21	104	
0103	RK09UDCS103		10,000	.15	122	
0104	RK09UDCS153		15,000	.13	150	
0105	RK09UDCS203		20,000 3/	.12	173	
0106	RK09UDCS253		25,000	.114	194	
0107	RK09UDCS353		35,000	.102	229	
0108	RK09UDCS503		50,000	.080	272	
0196	RK09UDCS753		75,000 4/	.060	290	
0197	RK09UDCS104		100,000 4/	.045	300	

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Standard bushing						
Dash number	Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal
0109	RK09SDCS100		10	.66	3.8	
0110	RK09SDCS150		15	.88	4.9	
0111	RK09SDCS200		20	.89	5.7	
0112	RK09SDCS250		25	.80	6.2	
0113	RK09SDCS350		35	.72	7.3	
0114	RK09SDCS500		50	.65	8.6	
0115	RK09SDCS750		75	.55	10	
0116	RK09SDCS101		100	.68	12	
0117	RK09SDCS151		150	.41	14	
0118	RK09SDCS201		200	.55	17.4	
0119	RK09SDCS251		250	.39	19	
0120	RK09SDCS351		350	.36	23	
0121	RK09SDCS501		500	.42	27	
0122	RK09SDCS751		750	.35	32	
0123	RK09SDCS102	.875	1,000	.34	39	S
0124	RK09SDCS152		1,500	.31	46	
0125	RK09SDCS202		2,000	.23	55	
0126	RK09SDCS252		2,500	.21	61	
0127	RK09SDCS352		3,500	.21	73	
0128	RK09SDCS502		5,000	.17	86	
0129	RK09SDCS752		7,500	.21	104	
0130	RK09SDCS103		10,000	.15	122	
0131	RK09SDCS153		15,000	.13	150	
0132	RK09SDCS203		20,000 3/	.12	173	
0133	RK09SDCS253		25,000	.114	194	
0134	RK09SDCS353		35,000	.102	229	
0135	RK09SDCS503		50,000	.080	272	
0198	RK09SDCS753		75,000 4/	.060	290	
0199	RK09SDCS104		100,000 4/	.045	300	

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Standard bushing						
Dash number	Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal
0136	RK09SGCS100		10	.66	3.8	
0137	RK09SGCS150		15	.88	4.9	
0138	RK09SGCS200		20	.89	5.7	
0139	RK09SGCS250		25	.80	6.2	
0140	RK09SGCS350		35	.72	7.3	
0141	RK09SGCS500		50	.65	8.6	
0142	RK09SGCS750		75	.55	10	
0143	RK09SGCS101		100	.68	12	
0144	RK09SGCS151		150	.41	14	
0145	RK09SGCS201		200	.55	17.4	
0146	RK09SGCS251		250	.39	19	
0147	RK09SGCS351		350	.36	23	
0148	RK09SGCS501		500	.42	27	
0149	RK09SGCS751		750	.35	32	
0150	RK09SGCS102	1.25	1,000	.34	39	S
0151	RK09SGCS152		1,500	.31	46	
0152	RK09SGCS202		2,000	.23	55	
0153	RK09SGCS252		2,500	.21	61	
0154	RK09SGCS352		3,500	.21	73	
0155	RK09SGCS502		5,000	.17	86	
0156	RK09SGCS752		7,500	.21	104	
0157	RK09SGCS103		10,000	.15	122	
0158	RK09SGCS153		15,000	.13	150	
0159	RK09SGCS203		20,000 3/	.12	173	
0160	RK09SGCS253		25,000	.114	194	
0161	RK09SGCS353		35,000	.102	229	
0162	RK09SGCS503		50,000	.080	272	
0200	RK09SGCS753		75,000 4/	.060	290	
0201	RK09SGCS104		100,000 4/	.045	300	

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Standard bushing						
Dash Number	Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal
0163	RK09SKCS100		10	.66	3.8	
0164	RK09SKCS150		15	.88	4.9	
0165	RK09SKCS200		20	.89	5.7	
0166	RK09SKCS250		25	.80	6.2	
0167	RK09SKCS350		35	.72	7.3	
0168	RK09SKCS500		50	.65	8.6	
0169	RK09SKCS750		75	.55	10	
0170	RK09SKCS101		100	.68	12	
0171	RK09SKCS151		150	.41	14	
0172	RK09SKCS201		200	.55	17.4	
0173	RK09SKCS251		250	.39	19	
0174	RK09SKCS351		350	.36	23	
0175	RK09SKCS501		500	.42	27	
0176	RK09SKCS751		750	.35	32	
0177	RK09SKCS102	2.500	1,000	.34	39	S
0178	RK09SKCS152		1,500	.31	46	
0179	RK09SKCS202		2,000	.23	55	
0180	RK09SKCS252		2,500	.21	61	
0181	RK09SKCS352		3,500	.21	73	
0182	RK09SKCS502		5,000	.17	86	
0183	RK09SKCS752		7,500	.21	104	
0184	RK09SKCS103		10,000	.15	122	
0185	RK09SKCS153		15,000	.13	150	
0186	RK09SKCS203		20,000 3/	.12	173	
0187	RK09SKCS253		25,000	.114	194	
0188	RK09SKCS353		35,000	.102	229	
0189	RK09SKCS503		50,000	.080	272	
0202	RK09SKCS753		75,000 4/	.060	290	
0203	RK09SKCS104		100,000 4/	.045	300	

1/ For reference purposes only.
2/ Percent of resolution = 1/N x 100. N = number of turns.
3/ Maximum value based on use of wire having no less than .001 inch (0.13 mm) nominal diameter.

NOTE: For Navy use, resistance values shall be based on the use of wire having no maximum less than 0.001 inch (0.03 mm) wire diameter.

4/ Valued based on use of wire having no less than 0.0006 inch (0.15 mm) nominal diameter.

NOTE: For Army and Air Force use, resistance values shall be based on the of wire having no less than 0.0006 inch (0.15 mm) wire diameter.

3.16 Marking. Due to size limitations, marking of the schematic diagram is optional.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-39002.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-39002 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this date of this specification, the applicable associated specification, and the complete PIN.
b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.
c. Packaging requirements (see 5.1).

6.3 Type designation. The type designation for identifying these parts is as follows:

NOTE: This is for information only. For the correct PIN, see 1.2.

RK09	SA	C	S	101
Style (6.3.1)	Shaft and type mounting (6.3.2)	Resistance temperature characteristic (6.3.3)	Terminal (6.3.4)	Resistance (6.3.5)

6.3.1 Style. The style is identified by the two letter symbol "RK" followed by a two digit number; the letters identify semi-precision wire-wound, variable resistors, and the number identified the physical size.

6.3.2 Shaft and type mounting. The style of shaft, type of mounting, and length of shaft is identified by a two letter symbol. The first letter indicates the style of shaft and type of mounting and the second letter indicates the length of shaft, see MIL-PRF-39002.

6.3.3 Resistance temperature characteristic. The resistance temperature characteristic of the completed resistor is identified by a single letter, see MIL-PRF-39002.

6.3.4 Terminals. The terminals are identified by a single letter, see MIL-PRF-39002.

6.3.5 Resistance. The nominal total resistance value expressed in ohms is identified by a three digit number; the first two digits represent significant figures and the last digit specified the number of zeros to follow, see MIL-PRF-39002.

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6.4 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.5 Amendment notification. The margins of this specification are marked with vertical lines to indicate modification generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship

Custodian:
Army - CR
DLA - CC

Preparing activity:
DLA - CC

Review activity:
Army - AR

(Project 5905-2018-035)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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